Fiscal Year:	FY 2005	**Task Last Updated:**	FY 05/04/2006
PI Name:	Weislogel, Mark M. Ph.D.		
Project Title:	Capillary Flow in Interior Corners		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Interfacial phenomena		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Project Type:	Flight	**Solicitation / Funding Source:**	98-HEDS-03
Start Date:	04/04/2002	**End Date:**	10/12/2004
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	3
No. of Bachelor's Candidates:	3	**Monitoring Center:**	NASA GRC
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Flight Program:			
Flight Assignment:	Note: PI changed organization/grant funded for flight		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NAG3-2741		
Performance Goal No.:			
Performance Goal Text:			
	Capillary flows in interior corners have an established place in fluids-handling operations in reduced gravity environments. A quantitative understanding of corner flows is essential for the myriad fluid management tasks in space including flows in liquid-fuel tanks, heat pipes, energy systems, condensers, and life-support systems. The applications-oriented study is both experimental and theoretical to understand capillary-driven flows in containers with interior corners as well as flows in complex containers corner networks. For the experimental investigation, drop tower and low-g aircraft experiments are performed to confirm theoretical predictions of transient 3-dimensional fluid processes fundamental to many in-flight fuels/propellant management and thermal control problems. For the complex corner network experiments, flows in planar and subsequently three-dimensional geometries of increasing complexity will be quantified in terms of corner geometry, size, fluid volume, and fluid properties. The overall goal of the proposed		

Task Description:	research is to provide a firm experimental and theoretical footing for the design of systems exploiting dymnamic capillary phenomena in interior corners, corner networks, and complex containers. Current techniques rely on order of magnitude estimates and costly computational modeling. Capillary flows in interior corners occur routinely in low-g fluid systems in space. Many such flows, if not anticipated, can have ruinous consequences. A quantitative understanding of corner flows is essential for the myriad fluid management tasks in space including flows in liquid-fuel tanks, heat pipes, energy systems, condensers, life-support systems, and biofluids processes. A quantitative understanding of such flows is a valuable contribution to the aerospace community. Such a contribution will aid in innovation of new designs, shorten design cycles, enhance system performance, and increase system reliability.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	Knowledge of capillary flows in interior corners is fundamental to many space fluids systems designs. Significant progress in the understanding of low-g corners flows was made during this effort. Specific problems including flows in arbitrary polygonal containers, stability, non-planar corners, transients, and steady flows with interfacial shear, mass and heat transfer have been solved. Extension of the results to complex corner networks has been made for steady flows. Flows in containers with curved corners and with dissimilar wettabilities have been analyzed. Many 2.2s and 5s drop tower and low-g aircraft tests were conducted to verify certain theoretical findings. Significant resources of this grant were used for NASA related outreach including support of three NASA Reduced Gravity Student Flight Opportunities Program (RGSOP) teams at Portland State University (PSU), and several high altitude balloon launches simulating satellite design and development. During the last two years of this contract (portion conducted at Portland State University) 1 Post-Doctoral Fellow, 1 M.S.M.E. student, 3 undergraduate M.E. students, and 5 high school summer interns through Oregon's Apprenticeships in Science in Engineering. Publications include 9 journal articles, 10 published presentations, and 15 other reports, invited presentations, etc. It was found that the analytical technique developed for capillary flow in an isolated corner has extensive applications in low-g fluid systems and that many accurate, closed form solutions to problems of immediate interest will continue to result. This report lists the accomplishments of this 4 year effort. The four years of this project have been productive and have made significant advances in the prediction of many capillary flows of practical importance to low-g fluid systems design: (1) imbibition in complex containers with interior corners, (2) flows in nonplanar corners, (3) steady flows with surface shear forces, heat and mass transfer, (4) steady flows in complex corner networks—a model of large scale flows in space and micro scale flows on Earth, and (5) a first identification of the aerospace surface software tool (Surface Evolver) to compute interface stability in complex containers. (6) The stability of wall-bound/edge-bound drops and bubbles was completed identifying a new critical wetting condition. A first analytical look at other important capillary driven corner flow problems was initiated: (6) flows in curved corners, (7) flows in rounded corners, (8) flows in corners with gaps, (9) flows in corners with wettability discontinuities, and transient flows in (10) tapered corners and (11) multi-corner junctions. The many corner flow problems addressed in the literature do not use the technique employed herein. Application of the technique to the problems of greatest historical importance will identify the subject technique as the correct approach to solving such problems. (The literature is replete with approximate hydraulic diameters, weighted capillary pressures, and friction factors, none of which need be invoked and many of which lead to errors of 30% or more.) As regards the steady corner flow research conducted to date, a single seminal publication is expected to eventually summarize this work. The corner flow solution approach developed herein should then be extended to unsteady corner flows in complex networks. Bifurcating and coalescing flows should be studied analytically and experimentally. Flows where capillary, viscous and inertial forces are of competing orders will be studied with the support of drop tower and aircraft experiments. All results useful in practical system design are being collected for a 'handbook' NASA contract report surveying capillary flows in complex geometric vessels, conduits, and tanks.
Bibliography Type:	Description: (Last Updated: 06/13/2025)
Articles in Peer-reviewed Journals	Collicott SH, Weislogel MM. "Computation of capillary instabilities using surface evolver." AIAA J. 2004 Feb;42(2):289-95. https://doi.org/10.2514/1.9093 , Feb-2004
Books/Book Chapters	Weislogel MM, Lichter S. "Water balloon rupture in low-g." in "Gallery of Fluid Motion." Ed. M. Samimy et al. Cambridge : New York : Cambridge University Press, 2004, p. 51., Jan-2004
Papers from Meeting Proceedings	Chen Y, Weislogel MM. "Analysis of Capillary Flow in Rounded Corners." Heat Transfer/Fluids Eng. Summer Conf., Charlotte, NC, July 11-15, 2004. Paper No. HT-FED2004-56253, July 2004 , Jul-2004
Papers from Meeting Proceedings	Weislogel MM, Sitorus AT. "Steady, Fully Developed Two-Phase Flow: A New Experimental Approach for Low-g Multiphase Flow Investigations." 34th AIAA Fluid Dynamics Conference, Portland, OR, June 28-July 1, 2004. Paper No. AIAA 2004-2644, June 2004. , Jun-2004
Significant Media Coverage	Weislogel MM. "Experiments to Analyze Fluids in Space." National Society of Professional Engineers, Engineering Times, Vol. 26, No. 3, March 2004., Mar-2004
Significant Media Coverage	Weislogel MM. "Web publication of Water Balloon Rupture in Low-g continues to build as well as email traffic." Web publication of Water Balloon Rupture in Low-g continues to build as well as email traffic. Well over 100,000 hits/visits recorded as of February 2004. February 2004, Feb-2004
Significant Media Coverage	Brooks Burford. "Preparations for Launch of PSU Experiment to ISS from Russia." KEX Radio Interview, January 26, 2004., Jan-2004